Runbook: Rotating TLS Certificates on the Edge Proxy Fleet

Owner	Platform Operations (Team Kestrel)
Last reviewed	2026-09-30
Severity if missed	SEV-2: browsers reject the expired certificate
Estimated time	45 minutes for all three pools

1. Purpose

This runbook replaces the public TLS certificate on every edge proxy before the current one expires. The certificate is renewed automatically, but installing it is still a manual, host-by-host operation. Follow the steps in order and do not skip the health check between hosts.

2. Before you start

- You hold the on-call role or have written approval from the on-call engineer.
- Your workstation can reach the bastion host and the secrets vault.
  - Test vault access with a read of a non-sensitive key.
  - Make sure your session token is valid for at least two hours.
- No other change is scheduled on the edge fleet in the same window.
- The renewed bundle is at least 24 hours old, so that every OCSP responder knows it.

> **Warning**
> Never reload more than one host of the same pool at a time. A pool with two hosts out of rotation cannot absorb a traffic peak, and the load balancer will start shedding requests.

3. Hosts in scope

Pool	Hosts	Region	Drain order	Notes
edge-fra	edge-fra-01 edge-fra-02 edge-fra-03	Frankfurt	1	Busiest pool; rotate outside 08:00–11:00 UTC
edge-dub	edge-dub-01 edge-dub-02	Dublin	2	edge-dub-02 also serves the status page
edge-mad	edge-mad-01 edge-mad-02	Madrid	3	None

4. Procedure

4.1 Preparation

1. Announce the change in the operations channel and open a change record.
2. Fetch the renewed bundle from the vault and verify its chain and expiry date:

```
vault kv get -field=bundle secret/edge/tls/2026 > /tmp/edge-2026.pem
openssl verify -CAfile /etc/ssl/internal-root.pem /tmp/edge-2026.pem
openssl x509 -in /tmp/edge-2026.pem -noout -enddate
```

4.2 Rolling the pools

3. For each pool, in the drain order of section 3, repeat for every host:

a. Remove the host from the load balancer and wait until fewer than 20 connections remain.
b. Copy the bundle to the host and reload the proxy.
c. Run the health check, then return the host to the load balancer.

For the first Frankfurt host the commands are:

```
edgectl drain edge-fra-01 --wait --max-conns 20
scp /tmp/edge-2026.pem edge-fra-01:/etc/edgeproxy/tls/current.pem
ssh edge-fra-01 'sudo systemctl reload edgeproxy'
edgectl health edge-fra-01 && edgectl undrain edge-fra-01
```

4.3 Verification

4. From a machine outside the corporate network, confirm that every pool serves the new expiry date.
5. Close the change record and post the new expiry date in the operations channel.

5. Configuration reference

The proxy reads the certificate path and the reload behaviour from the following section of its configuration file.

```
tls:
  certificate: /etc/edgeproxy/tls/current.pem
  min_version: "1.2"
  ciphers:
    - TLS_AES_128_GCM_SHA256
    - TLS_CHACHA20_POLY1305_SHA256
  ocsp_stapling: true
reload:
  grace_period: 30s
  drain_timeout: 120s
```

A reload keeps existing connections open for the grace period, so clients finish their requests on the old certificate. Do not shorten the drain timeout during a rotation.

6. Rollback

If a host fails its health check after the reload, stop the rotation and roll that host back.

R1. Copy the previous bundle, kept as previous.pem on every host, over current.pem.

R2. Reload the proxy and run the health check again.

R3. Leave the host drained if the check still fails, and page the network team.

7. Escalation

Time since start	Action
30 minutes	Tell the on-call lead that the rotation is running late.
60 minutes	Page the network team and pause the rotation.
Expiry within 24 hours	Open a SEV-2 incident and follow the incident process.